



SEED SIGNER



Summer of Bitcoin 2025

Project Information:

Name: SeedSigner Enhancement - SeedXOR ([here](#))

Difficulty: Medium

Mentor: [@kdmukai](#) , [@SeedSigner](#)

Student Details:

Full Name: Advait Gaur

Email: advaitgaur2003@gmail.com

GitHub: [advaitgaur004](#)

LinkedIn: [Advait Gaur](#)

Twitter: [Advait Gaur](#)

Preferred Language: I'm proficient in English for communication, both spoken and written

Location: Currently Jodhpur (Rajasthan), India

Time Zone: Indian Standard Time (IST) (UTC +05:30)

Phone Number: +91 8854042218

Institution: IIT Jodhpur

Program: B. Tech in Computer Science and Engineering

Stage of completion: 3rd Year (college completion expected June 2026)

Open-Source Contribution

Contribution to Seed-Signer ❤️

- [\[Fix\]](#) Update capitalization to Sentence case for buttons
- [\[Fix\]](#) seed pagination logic in SeedWordsView
- [\[Tool\]](#) Add SeedSigner XOR Seed Backup Tool for splitting and combining seed phrases
- [\[Refactor\]](#) Fix D-pad and hardware key alignment in IOTestScreen
- [\[Enhanced\]](#) seed management with BIP-85 support
- [\[Feature\]](#) Seed XOR functionality in Seeds Menu

Open
Merged
Draft

Other Open-Source Competition Contributions

- TRS-Archive (under DevUp Labs)
 - I got [21 Merged PRs](#). I was the top contributor to the project.
- Arvix-devs
 - Add mode method for DiscreteWeibull, LogitNormal, Hypergeometric, and SkewNormal ([#636](#))
 - Add consistent legend support across backends ([#115](#))
 - Enhance pre-commit configuration with additional hooks and language Version ([#22](#))
- CCExtractor
 - Add Support for Tags in Task Management ([#74](#))
 - Add Rate Limiting to API Endpoints ([#76](#))
 - [\(BACKEND\)](#) - Add recurrence support for tasks ([#93](#))
- Served as a co-mentor for a [SWOC](#) 24' project.

I have participated in many hackathons like **Opencode'23**, **DUHacks 3.0**, **Hackprints 3.0**, **FusionHackZ**, **HackNITR 5.0**, **SWOC Season 5**, and many others (I've lost count). I've built various projects ranging from AI/ML to Web Development, Blockchain, Cloud Computing, and Game Development. **Currently, I am the coordinator of the [DevUp Labs](#) (Maintainer of the org) and mentoring 1200+ Juniors**

- **Bug Bounty (one of the projects under My Guidance)** 🦋

This is more than just a project; it's a program by DevUp Labs under *my leadership* to encourage freshers to participate in open source. We've provided them with an [issue tracker](#) for projects under DevUp Labs for collaboration and growth as numerous people start working on these projects (within a single day, seven issues were resolved), helping each other along the way. 😊

Why Seed Signer?

I'm excited about SeedSigner because it builds tools that keep Bitcoin seeds safe and easy to use without trusting anyone else. Its open-source project lets people control their money with a simple, offline device. I love how SeedSigner uses clever ideas like SeedXOR and coin flips to make super secure seeds, which matches my interest in keeping things safe and private.

SeedSigner's focus on key management is why I'm drawn to it. I'm particularly interested in implementing SeedXOR functionality, which allows users to combine multiple seed phrases using XOR operations. This provides advanced security options like seed splitting or combining multiple entropy sources, giving users more flexibility in their security models without sacrificing the core principles of air-gapped operations. The project moves fast, with new updates all the time, like better screens and stronger security. I am connected to this because I want to grow with a project changing how people use Bitcoin. **The mentors, even though they're volunteers, are excellent.** They help newcomers like me understand tough stuff and make me feel part of the team. I want to join SeedSigner now and stick around after the coding phase to see where this fantastic project goes next.

Presence in the Community:

As an open-source contributor and maintainer of an organization, I understand the importance of effective communication with mentors and co-mentors on respective projects. I actively participate in telegram discussions, seeking suggestions and approvals before proceeding with fixes. I have also had in-depth conversations about the project with seniors like Keith, Seedsigner, Jean-Do, and Dimi (sometimes 😊).

One such example is the discussion about SeedXOR functionality. There was an informative thread where community members debated the merits of XOR versus other schemes like Shamir's Secret Sharing. As I correctly questioned in the Telegram discussion, *"If splitting a Bitcoin seed using XOR makes it impossible to recover the seed from just one part, why might someone choose XOR splitting over Shamir's Secret Sharing, which allows recovery from a subset of parts."*

The response highlighted that *"XOR splitting stands out when a stricter security model is desired, offering the added benefit of being simpler,"* while *"Shamir's Secret Sharing is better suited when threshold flexibility is important."* These insights show that both approaches have their place, and SeedXOR fills an important niche in the ecosystem.

Apart from that, I love to [review some PRs](#) that other contributors raise; it helps us get to know the codebase even more. I always put as much effort into the community so that it becomes better for me and new people. Now, I've made some friends in the organization (new contributors) who share a similar mindset.

Implementation

Now, this is the most interesting part of the entire proposal.

SeedSigner Enhancement – SeedXOR

Objective:

Enhance SeedSigner by adding a SeedXOR feature that allows users to combine multiple seed phrases (or their derived keys) using the XOR operation. This will enable advanced key management scenarios while maintaining compatibility with BIP-39 standards and SeedSigner's security principles.

Background:

SeedXOR is a method to securely split or merge cryptographic seeds. By performing an XOR operation on multiple seeds, users can create a combined seed that can be used for various purposes, such as creating a backup that requires multiple parts to recover the original seed.



Current Implementation:

For Detailed Discussion, one can Look into the Issue and Base Implementation by [Jean Do](#) and this can be carried forward

- [Feature] Add SeedXOR implementation ([Issue #43](#))
- [Feature SeedXOR] **I am Already Working on this PR** ([PR#738](#))

Presently, SeedSigner does not have built-in SeedXOR capabilities. Users seeking this functionality must currently perform these operations manually or use external tools, which is inconvenient and potentially introduces security risks if done incorrectly.

Problem Statement

This project aims to implement SeedXOR functionality in SeedSigner, enabling users to:

1. **Split an existing seed into multiple shares using XOR**
2. **Combine multiple seeds to create a new seed**
3. Integrate these operations seamlessly into SeedSigner's user interface
4. Ensure all operations maintain compatibility with BIP-39 standards
5. Implement a "**Seed Vault**" (***optional*) concept for storing intermediary XOR results

Proposed Change

Core XOR Functionality:

- Implement XOR operations for seed data in [src/seedsigner/models/seed.py](#)
- Support combining multiple component keys (2 or more) via XOR
- Ensure compatibility with various seed lengths (12/24 words)
- Add validation to ensure outputs are valid BIP-39 seeds
- Design temporary component key storage that discards keys after final combination

Workflow Integration:

- Add "Rebuild SeedXOR" option within the existing "Load A Seed" flow
- Implement sequential loading of component keys:
 - First component key via SeedQR or mnemonic entry
 - Additional component keys via the same methods
 - Clear indication of progress (e.g., "Key 1 of N loaded")
- Display SeedXOR fingerprint after each key combination
- Provide a decision point after each combination:
 - Load another component key
 - Proceed to finalization options

Finalization Options:

- Create a finalization screen with options:
 - Add BIP-39 passphrase
 - Complete process (no passphrase)
- Result: Only store the final XORed key in memory
- Automatically discard all individual component keys after combination

Security and UX Considerations:

- Clear user guidance throughout the process
- Progress indicators showing how many keys have been combined
- Visual differentiation between component keys and the resulting key
- Warnings about the cryptographic properties of XOR operations
- Confirmation steps to prevent accidental data loss
- Enable users to cancel the process at any point

Testing and Documentation:

- Test cases for various combinations (2, 3, 4+ component keys)
- Edge case testing for different seed lengths
- User documentation explaining SeedXOR usage scenarios
- Developer documentation for the implementation
- Update screenshot generator to include new UI screens

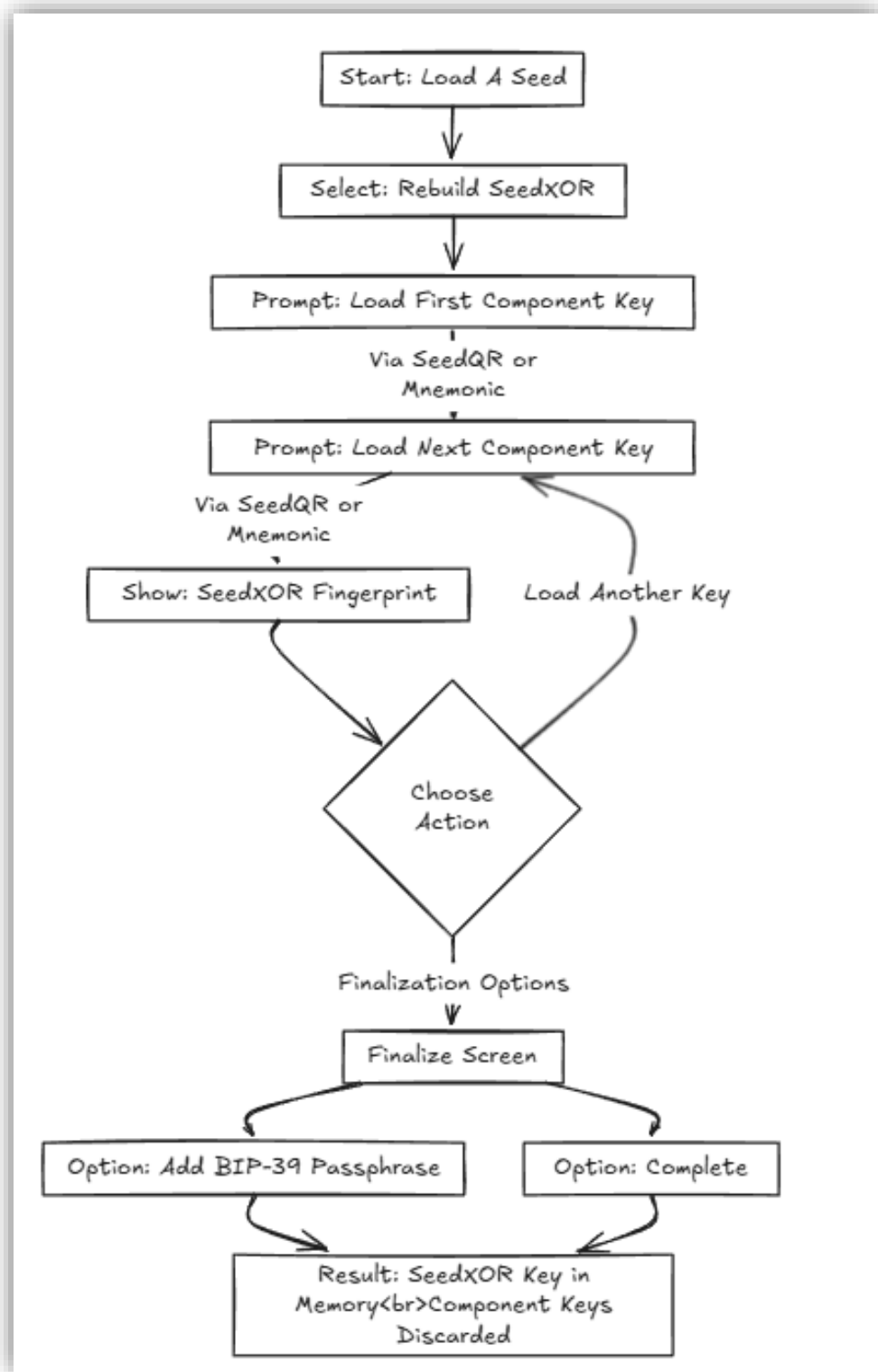
This implementation follows a flow-based philosophy where the user first selects their intention (rebuild a seed using XOR), then follows a guided process to load and combine component keys until they reach their desired result. The focus is on a streamlined user experience that maintains security while reducing complexity for the user.

Seed Vault Concept (**Optional Change)

I propose adding a "Seed Vault" feature to SeedSigner's SeedXOR implementation. This would allow:

1. **Temporary seed storage:** Users can store intermediate XOR results temporarily on the device
2. **Step-by-step combination:** Add multiple XOR parts one by one without needing to have them all at once
3. **Enhanced privacy:** Users can travel with individual XOR parts and combine them only when needed
4. **Secure cleanup:** Options to wipe vault seeds after use to maintain security

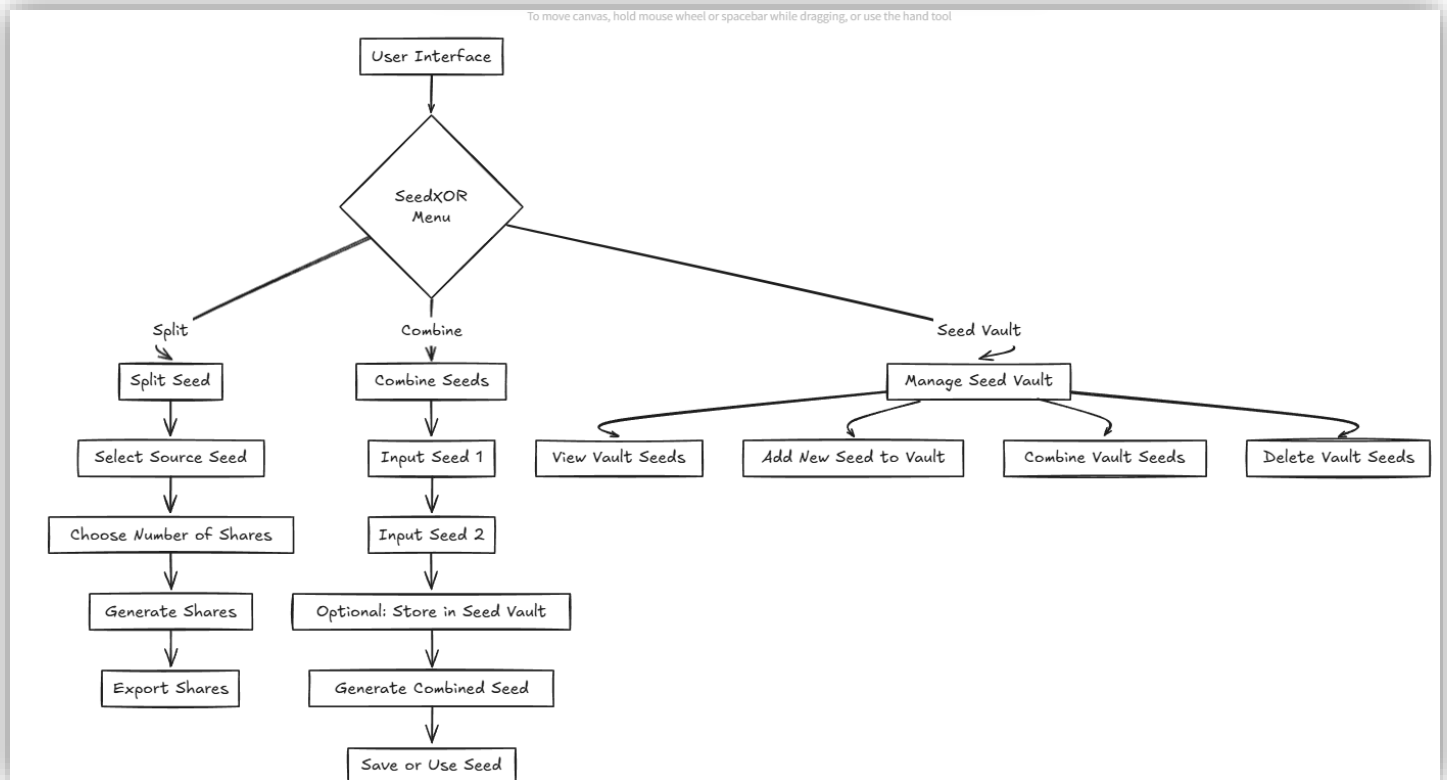
Implementation Diagram:



(If the diagram appears blurry, open the [link](#) directly for better clarity.)

I created this flow chart after discussing the idea with my mentor on Telegram, and finally shared the result on GitHub. ([Click Here](#))

The Seed Vault would be implemented as a memory-only feature (cleared on device restart) to maintain Seed Signer's stateless security model, with an option for users to export a QR code of the intermediate result if needed for backup. (**Optional Feature)

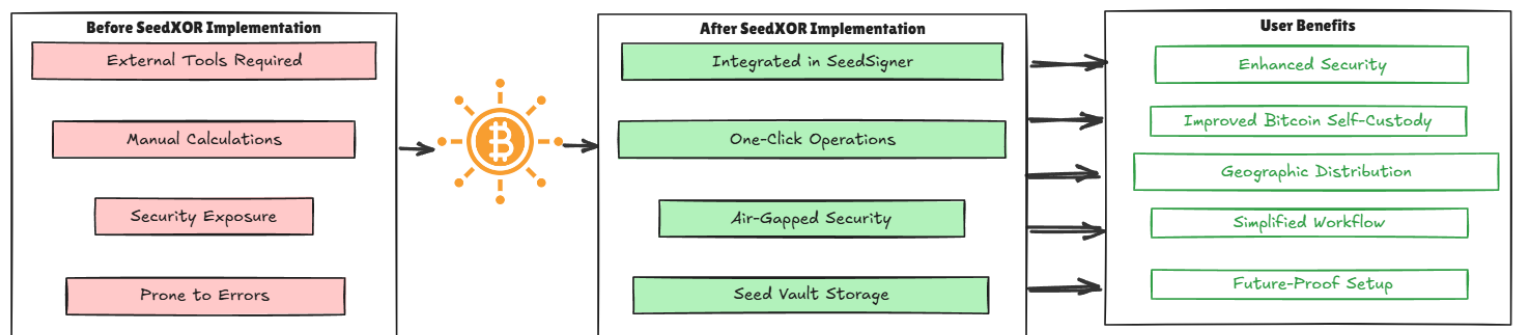


(If the diagram appears blurry, open the [link](#) directly for better clarity.)

Before and After

I'll create a cute diagram on Excalidraw to illustrate the difference before and after the full implementation of SeedXor in SeedSigner.

(If the diagram appears blurry, open the [link](#) directly for better clarity.)



Timeline:

Time Frame	Tasks	Priority (Out of 5)
Coding officially begins! May 15 – May 21	Initial setup, team alignment, and planning with mentors.	
May 22 – May 28	Research XOR operations for seeds and draft implementation approach, on the (PR#738)	
May 29 – June 4	Implement core XOR functions in seed.py for both seed splitting and combining	
June 5 - June 11	Create UI screens for SeedXOR operations and initial integration.	
June 12 – June 18	Implement seed validation, error handling, and edge cases	
June 19 - June 30	Run initial tests, share progress with mentor for review, and adjust based on input.	
Midterm evaluation deadline July 1 - July 5	Mid-Term Evaluation Submission	
July 6 – July 16	Refine UI, add warning screens and confirmations	
July 17 - July 23	Enhance screenshot generator with new SeedXOR UI screens	
July 24 - Aug 15	Conduct comprehensive testing and fix any bugs.	
Students Submit Projects Aug 15	Students submit a final project report, including project summaries and links to code submissions or artifacts.	

About Me

I believe in showing code, rather than talking myself up. Actions speak louder than words in the open-source world. I invite the admins and mentors to understand who I am by looking at my code, the true reflection of a developer's passion, thinking, and dedication. I've always followed a project-based approach while learning new technologies. This hands-on philosophy has led me to build countless projects. Here are some of them:

Project	Description	Link
TRS-Archive	Technical review system for academic submissions in robotics, (Replica of Arxiv exclusively for Robotics Society)	SRS Codebase
Ignus 25	Web app for IIT Jodhpur's Socio-Cultural fest serving 16,000+ users	Live Codebase
Bitcoin Explorer	Tool for BTC balance checks, transaction history, and live conversion	GitHub
Whispernet	P2P network using Gossip protocol for message propagation, featuring real-time liveness detection, deduplication, and dynamic peer management	GitHub
SLR (Sign Language Recognition)	Project using CNN, MediaPipe, and Transfer Learning for sign language recognition	GitHub
Mini DB	Lightweight C++ database with SQL-like operations and ACID compliance	Github Demo
OS Error Handler	System with restricted log file permissions and resource monitoring	GitHub
Auto Capture Selfie	Real-time gesture detection using EfficientDet-D0 with Kotlin and TFLite	SRS Pipeline Github
Movie Recommendation System	Hybrid filtering with neural networks for personalized movie recommendations	Live Abstract Demo

Why Me!?

As a maintainer of an organization, I understand that open source thrives on trust and consistency. I've been contributing to DevUp Labs for over a year, and this consistent presence is what builds a true sense of belonging within open-source communities, where people selflessly support one another.

For Summer of Bitcoin 2025, my primary goal is to work alongside amazing people under mentorship, fuelling my own growth while motivating junior developers to keep pushing forward, a continuous cycle of inspiration. **I'm committed to contributing even after the summer term ends because I genuinely love open source.** 😊

Availability:

I plan to dedicate 30-50 hours per week to the project and will be most active Any time (workaholic) from 11 AM to 11 PM IST (except for 5-7 PM workout). I will have my End-semester exams between 20 April - 26 April. So, I am free after that, so I can actively participate during the *Community Bonding Period* and enjoy the coding period 😊.

Conclusion:

Lastly, I would say that **I am confident I can complete this project within the given timeline and take full responsibility for implementing all the crucial and valuable features** that will take SeedSigner to the next level.

I am 100% dedicated to SeedSigner and have no plans whatsoever to submit a proposal to any other organization.